

Olist E-Commerce — Business Intelligence Report

Brazilian marketplace dataset · 2016–2018 · Google BigQuery (GoogleSQL)

EXECUTIVE SNAPSHOT

Total late orders	Avg delivery (overall)	Avg delivery (late)
6,534	12 days	33 days
Across 2016–2018	All delivered orders	2.75× above average

Avg logistics time	Avg dispatch time	Peak late-order month
25 days	5 days	Mar 2018
Carrier → customer (late)	Purchase → carrier (late)	1,328 orders · ~20% of total

MONTHLY LATE ORDER DISTRIBUTION

Month	Year	Late Orders
Nov	2017	904
Dec	2017	411
Jan	2018	403
Feb	2018	926
Mar	2018	1,328
Apr	2018	306
May	2018	443
Jun	2018	71
Jul	2018	208
Aug	2018	393
Total (10-month window)		5,393

Rows highlighted in red = months with 400+ late orders.

LATE DELIVERY ANALYSIS

Over the three-year period (2016–2018), a total of **6,534 orders were delivered past their estimated date** — 3 in 2016, 2,453 in 2017, and 4,078 in 2018, indicating a sharp upward trajectory. Within the 10-month window analysed, 7 of 10 months recorded 400+ late orders. The situation peaked in **March 2018 with 1,328 late deliveries**, representing approximately 20% of all late orders across the entire three-year dataset. This concentration points to systemic pressures — likely a combination of **dispatch delays averaging 5 days** before carrier handoff and **in-transit logistics averaging 25 days** for late orders — compared to the 12-day average for on-time deliveries. Addressing dispatch scheduling and last-mile carrier capacity during Q1 demand surges would be the highest-impact intervention.

KEY ANALYTICAL FINDINGS

- **CLV segmentation via NTILE:** Customers ranked into HIGH / MID / LOW tiers using a weighted composite score (60% total spend + 40% order frequency), enabling targeted retention and upsell strategies.
- **State-level delivery performance:** Average delivery time calculated per Brazilian state using `TIMESTAMP_DIFF` — identifies geographic bottlenecks for logistics prioritisation.
- **Top revenue categories:** Top 5 product categories ranked by total payment value (in millions BRL) — health & beauty and watches/gifts consistently lead revenue contribution.
- **6-month repeat purchase rate:** `LAG()` window function used to identify customers who re-ordered within 180 days — a critical retention KPI for marketplace health.
- **High-volume seller quality:** Sellers with 50+ delivered orders ranked by average review score using a deduplicated CTE, surfacing reliable, high-performing marketplace partners.

Tools: Google BigQuery · GoogleSQL · Window functions · CTEs · `TIMESTAMP_DIFF` · `NTILE` · `LAG`